

National University

Of Computer & Emerging Sciences Faisalabad - Chiniot Campus

CS 1005 : Discrete Structures

Assignment 5

Instructor : Ms. Mahzaib Younas

Section BCS (C)

Q No 01: Let $G = \{V, E\}$ be a graph. Draw a graph with the following specified properties, or explain why no such graph exists. A graph having

$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

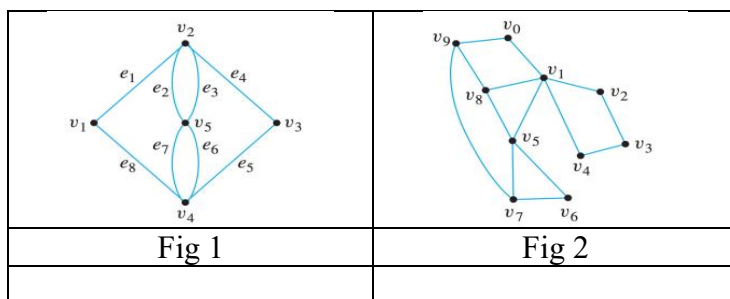
$$E = \{e_1, e_2, e_3, e_4, e_5\},$$

where $e_1 = (v_1, v_2)$, $e_2 = (v_2, v_4)$, $e_3 = (v_1, v_4)$, $e_4 = (v_3, v_3)$ and $e_5 = (v_3, v_5)$

Q No 02: Either draw a graph with the following specified properties, or explain why no such graph exists:

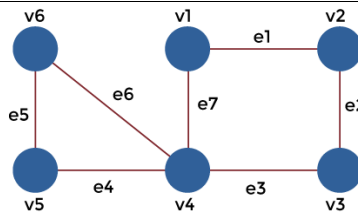
1. A graph with four vertices having the degrees of its vertices 1, 2, 3 and 4.
2. A graph with five edges having the degrees of its vertices 1, 1, 3 and 3.
3. A graph with four vertices having the degrees of its vertices 1, 1, 2 and 2

Q No 03: Check if the graphs in Fig. 1 and Fig. 2 have an Euler circuit. If the graph does not have an Euler circuit, explain why not. If it does have an Euler circuit, describe one.



Q No 04: In the graph given below, With the help of below sequences, you have to determine that the given sequence is walk, closed walk, circuit, closed circuit, path or closed path

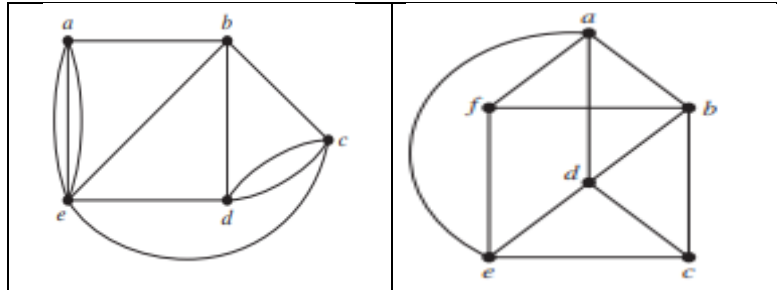
1. $v_1, e_1, v_2, e_2, v_3, e_2, v_2$
2. $v_4, e_7, v_1, e_1, v_2, e_2, v_3, e_3, v_4, e_4, v_5$
3. $v_1, e_1, v_2, e_2, v_3, e_3, v_4, e_4, v_5$
4. $v_1, e_1, v_2, e_2, v_3, e_3, v_4, e_7, v_1$
5. $v_6, e_5, v_5, e_4, v_4, e_3, v_3, e_2, v_2, e_1, v_1, e_7, v_4, e_6, v_6$



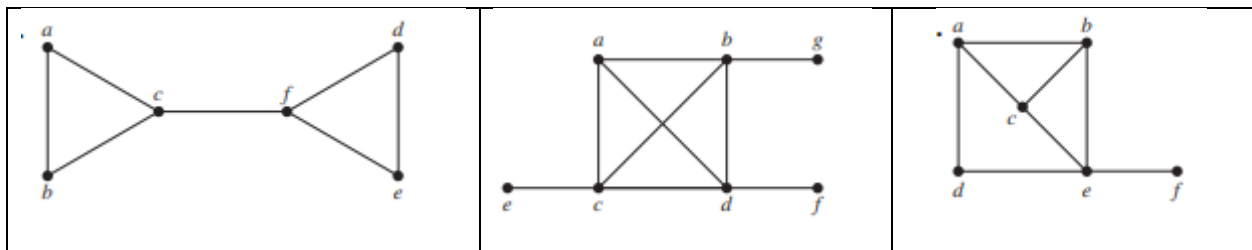
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Q No 05: Determine whether the given graph has an Euler circuit. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has an Euler path and construct such a path if one exists

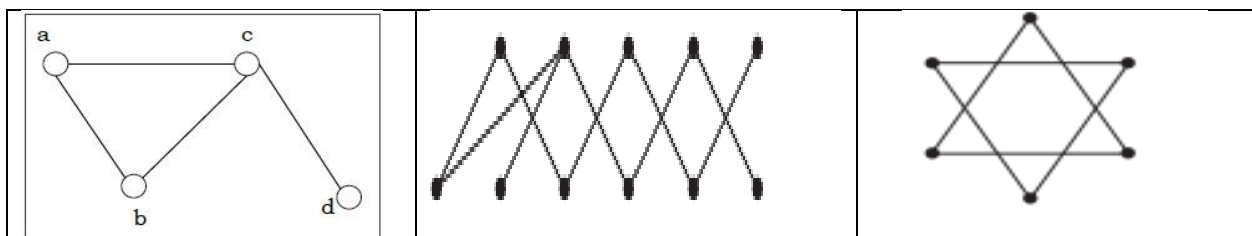


Q No 06: Write the properties of Hamiltonian circuit and Determine whether the given graph has a Hamilton circuit. If it does, find such a circuit. If it does not, give an argument to show why no such circuit exists.



Q No 07: Draw two 4-regular graphs.

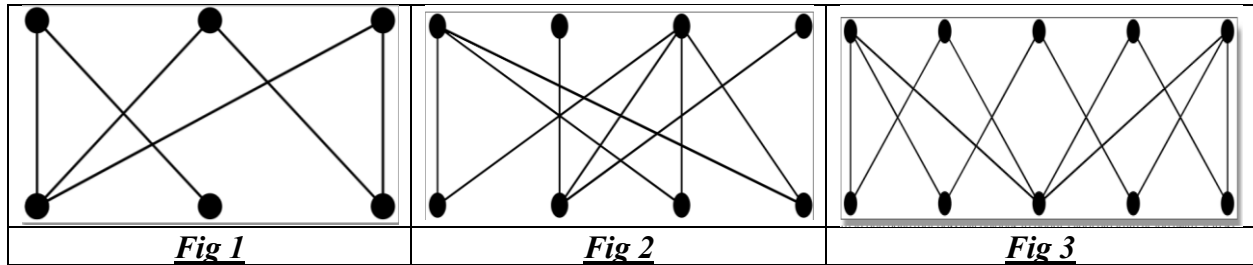
Q No 08: Determine whether the given graph is connected or not.



Q No 09: Find a matching of the bipartite graphs below or explain why no matching exists.

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Q No 10: Write down adjacency matrix for each vertex of given graphs

